

Exercise for Sect. 2.7

Exercise

2.7.1

Show that if a measure-preserving system $(X, \mathcal{B}, \mu, T)$ has the property that for any $A, B \in \mathcal{B}$ there exists N such that

$$\mu(A \cap T^{-n}B) = \mu(A)\mu(B)$$

for all $n \geq N$, then it is trivial in the sense that $\mu(A) = 0$ or 1 for every $A \in \mathcal{B}$.

Proof: If $\mu(A) = 0$ or 1 , the equation is trivial true for all $n \in \mathbb{N}$. Thus, suppose $A \in \mathcal{B}$ that $\mu(A) = c \in (0, 1)$,

$$\mu(A \cap T^{-n}A) = \mu(A)\mu(A) = c^2$$

for some $N_0 \in \mathbb{N}$ and all $n \geq N_0$. Choose an subsequence $\{A_{n_j}\}$ of $\{A_n = T^{-n}A\}$ such that $\mu(A_{n_j} \cap A_{n_k}) = \mu(A_{n_j})\mu(A_{n_k}) = c^2$ by letting $A_{n_0} = A_0 = A$, $n_1 \geq N_0$, $n_2 \geq N_1$ that $\mu(A_1 \cap T^{-n}A) = \mu(A_1)\mu(A)$ for all $n \geq N_2$. Thus, $\mu(A_{n_0} \cap A_{n_2}) = \mu(A_{n_1} \cap A_{n_2}) = c^2$ and then choose n_3, n_4 and so on.

□

Exercise

2.7.2

Show that if a measure-preserving system $(X, \mathcal{B}, \mu, T)$ has the property that

$$\mu(A \cap T^{-n}B) \rightarrow \mu(A)\mu(B)$$

uniformly as $n \rightarrow \infty$ for every measurable $A \subseteq B \in \mathcal{B}$, then it is trivial in the sense that $\mu(A) = 0$ or 1 for every $A \in \mathcal{B}$.

Proof: Suppose $B \in \mathcal{B}$ and for all $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that $|\mu(A \cap T^{-n}B) - \mu(A)\mu(B)| < \varepsilon$ for all $n \geq N$. Let $A_n = B \cap T^{-n}B$ and we have $A_n \cap T^{-n}B = B \cap T^{-n}B \cap T^{-n}B = A_n$. Hence for $n \geq N$,

$$|\mu(A_n)(1 - \mu(B))| = |\mu(A_n) - \mu(A_n)\mu(B)| = |\mu(A_n \cap T^{-n}B) - \mu(A_n)\mu(B)| < \varepsilon.$$

Apply the equation again,

$$|\mu(A_n) - \mu(B)^2| = |\mu(A \cap T^{-n}B) - \mu(B)\mu(B)| < \varepsilon.$$

Thus,

$$|\mu(B)^2(1 - \mu(B))| \leq |\mu(B)^2(1 - \mu(B)) - \mu(A_n)(1 - \mu(B))| + |\mu(A_n)(1 - \mu(B))| < 2\varepsilon.$$

Since $\mu(B) \in [0, 1]$ and ε is arbitrary, we have $\mu(B)^2 - \mu(B)^3 = 0$ and thus $\mu(B) \in \{0, 1\}$.

□

Exercise

2.7.3

Assume that $\mathcal{A}$ is a semi-algebra that generates $\mathcal{B}$, and prove the following characterizations of the basic mixing properties for a measure-preserving system $(X, \mathcal{B}, \mu, T)$:

1. T is mixing if and only if

$$\mu(A \cap T^{-n}B) \rightarrow \mu(A)\mu(B)$$

as $n \rightarrow \infty$ for all $A, B \in \mathcal{A}$.

2. T is weak-mixing if and only if

$$\frac{1}{N} \sum_{n=0}^{N-1} |\mu(A \cap T^{-n}B) - \mu(A)\mu(B)| \rightarrow 0$$

as $N \rightarrow \infty$ for all $A, B \in \mathcal{A}$.

3. T is ergodic if and only if

$$\frac{1}{N} \sum_{n=0}^{N-1} \mu(A \cap T^{-n}B) \rightarrow \mu(A)\mu(B)$$

as $N \rightarrow \infty$ for all $A, B \in \mathcal{A}$.

Proof: The $\Rightarrow$ sides are trivial, so just prove the $\Leftarrow$ sides.

1. Suppose $A, B \in \mathcal{B}$ and $\varepsilon > 0$, there exists $\{A_j\}_{j=1}^r$ and $\{B_k\}_{k=1}^l$ that

$$\mu\left(A \Delta \left(\bigcup_{j=1}^r A_j\right)\right) < \varepsilon \text{ and } \mu\left(B \Delta \left(\bigcup_{k=1}^l B_k\right)\right) < \varepsilon.$$

□

Exercise

2.7.4

Let $\mathcal{A}$ be a generating semi-algebra in $\mathcal{B}$ and assume that for $A \in \mathcal{A}$,
 $\mu(A \Delta T^{-1}A) = 0$ implies $\mu(A) = 0$ or 1 . Does it follow that T is ergodic?

Proof: Consider $X = \mathbb{T}^2$, $\mathcal{A}$ is the set of all rectangles and μ is Lebesgue measure. $T = R_\alpha \times R_\alpha$ where $R_\alpha : \mathbb{T} \rightarrow \mathbb{T}$ that $R_\alpha x = x + \alpha \bmod 1$ with α is irrational. T is not ergodic but satisfies the assumption. □

Exercise

2.7.5

Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is mixing if and only if

$$\lim_{n \rightarrow \infty} \langle U_T^n f, g \rangle = \langle f, 1 \rangle \langle 1, g \rangle$$

for all f and g lying in a dense subset of L^2_μ .

Proof:

□

Exercise

2.7.6

Use exercise 2.7.5 to prove the following.

1. An ergodic automorphism of a compact abelian group is mixing with respect to Haar measure.
2. An ergodic automorphism of a compact abelian group is mixing of all orders with respect to Haar measure.

Proof:

□

Exercise

2.7.7

Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is weak-mixing if and only if

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |\langle U_T^n f, g \rangle - \langle f, 1 \rangle \langle 1, g \rangle| = 0$$

for any $f, g \in L^2_\mu$.

Proof:

□

Exercise**2.7.8**

Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is weak-mixing if and only if

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |\langle U_T^n f, f \rangle - \langle f, 1 \rangle \langle 1, f \rangle| = 0$$

for any $f \in L^2_\mu$.

Proof:

□

Exercise

2.7.9

Show that a Bernoulli shift is mixing of order k for every $k \geq 1$.

Proof:

□

Exercise**2.7.10**

Prove that a measure-preserving transformation T is mixing if and only if

$$\mu(A \cap T^{-n}A) \rightarrow \mu(A)^2$$

for all $A \in \mathcal{B}$. Deduce that T is mixing if and only if $\langle U_T^n f, f \rangle \rightarrow 0$ as $n \rightarrow \infty$ for all f in a set of function dense in the set of all L^2 functions of zero integral.

Proof:

□

Exercise**2.7.11**

Prove that a measure-preserving transformation T is weak mixing if and only if for any measurable sets A, B, C with positive measure, there exists some $n \geq 1$ such that $T^{-n}A \cap B \neq \emptyset$ and $T^{-n}A \cap C \neq \emptyset$.

Proof:

□

Exercise**2.7.12**

Write T^k for the k -fold Cartesian product $T \times \cdots \times T$. Prove that T^k is ergodic for all $k \geq 2$ if and only if T^2 is ergodic.

Proof:

□

Exercise**2.7.13**

Let T be an ergodic endomorphism of $\mathbb{T}^d$. The following exponential error rate for the mixing property,

$$\left| \langle f_1, U_T^n f_2 \rangle - \int f_1 \int f_2 \right| \leq S(f_1)S(f_2)\theta^n$$

for some $\theta < 1$ depending on T and for the pair of constants $S(f_1), S(f_2)$ depending on $f_1, f_2 \in C^\infty(\mathbb{T}^d)$, is known to hold.

1. Prove an exponential rate of mixing for the map $T_n : \mathbb{T} \rightarrow \mathbb{T}$ defined by

$$T_n(x) = nx \bmod 1.$$

2. Prove an exponential rate of mixing for the automorphism of $\mathbb{T}^2$ defined by

$$T : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} y \\ x+y \end{pmatrix}.$$

3. Could an exponential rate of mixing hold for all continuous functions?

Proof:

□